

Exponential damping of a bounded semigroup does not require K -convexity

Solution packet for arXiv:1907.03113, Question 5.8

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Candidate full answer. Question 5.8 has an affirmative answer on every Banach space. In fact, for a bounded semigroup, local γ -boundedness, one exponentially damped γ -bounded orbit, and all exponentially damped γ -bounded orbits are equivalent.

1 The question

The source is Loris Arnold, *γ -boundedness of C_0 -semigroups and their H^∞ -functional calculi*, arXiv:1907.03113v1 [1]. On PDF page 20 it asks:

Question 5.8. Does the assumption X is K -convex space in Corollary 5.5 can be dropped?

Let X be a Banach space and $(T_t)_{t \geq 0}$ a bounded C_0 -semigroup on X . Corollary 5.5 of the source proves, under the assumption that X is K -convex, that

$$\{e^{-\delta_0 t} T_t : t \geq 0\} \text{ is } \gamma\text{-bounded for one } \delta_0 > 0 \\ \implies \{e^{-\delta t} T_t : t \geq 0\} \text{ is } \gamma\text{-bounded for every } \delta > 0.$$

Question 5.8 asks whether K -convexity can be dropped. We prove that it can. The proof uses only the semigroup law, ordinary boundedness, and two elementary facts about γ -bounded families.

For a family $\mathcal{T} \subset \mathcal{L}(X)$, write $\gamma(\mathcal{T})$ for its γ -bound, namely the least C such that

$$\left\| \sum_{j=1}^N \gamma_j S_j x_j \right\|_{L^2(\Omega; X)} \leq C \left\| \sum_{j=1}^N \gamma_j x_j \right\|_{L^2(\Omega; X)}$$

for all finite choices $S_j \in \mathcal{T}$ and $x_j \in X$, where the γ_j are independent complex standard Gaussian variables.

2 Proof intuition

Exponential damping makes the long-time part of a bounded semigroup summable. Split time into its integer and fractional parts, $t = n + s$. The fractional-time operators lie in one fixed local family, which inherits γ -boundedness from the single assumed damped orbit. Meanwhile the integer-time operators $e^{-\delta n} T_n$ have summable norms. A norm-summable operator family is γ -bounded even when selected indices repeat: group equal indices and view each Gaussian block as a conditional expectation of the full Gaussian sum. Multiplying the discrete and local families then reconstructs every $e^{-\delta t} T_t$. This is precisely where the source's global geometric assumption becomes unnecessary.

3 Two elementary lemmas

Lemma 1 (summable families). *Let $(A_n)_{n \geq 0} \subset \mathcal{L}(X)$ satisfy $\sum_{n \geq 0} \|A_n\| < \infty$. Then $\{A_n : n \geq 0\}$ is γ -bounded and*

$$\gamma(\{A_n : n \geq 0\}) \leq \sum_{n \geq 0} \|A_n\|.$$

Proof. Fix finite choices $n_1, \dots, n_N$ and $x_1, \dots, x_N \in X$. The selected indices may repeat, which is the only point requiring care. Put $I_n = \{j : n_j = n\}$ and $Z = \sum_{j=1}^N \gamma_j x_j$. Conditional expectation onto the Gaussian coordinates with indices in I_n gives

$$P_n Z := \mathbb{E}[Z \mid \sigma(\gamma_j : j \in I_n)] = \sum_{j \in I_n} \gamma_j x_j, \quad \|P_n Z\|_{L^2(\Omega; X)} \leq \|Z\|_{L^2(\Omega; X)}.$$

Consequently,

$$\begin{aligned} \left\| \sum_{j=1}^N \gamma_j A_{n_j} x_j \right\|_{L^2(\Omega; X)} &= \left\| \sum_n A_n P_n Z \right\|_{L^2(\Omega; X)} \\ &\leq \sum_n \|A_n\| \|P_n Z\|_{L^2(\Omega; X)} \\ &\leq \left(\sum_n \|A_n\| \right) \|Z\|_{L^2(\Omega; X)}. \end{aligned}$$

Only finitely many $P_n Z$ are nonzero, so all displayed sums are legitimate. $\square$

Lemma 2 (products). *If $\mathcal{A}, \mathcal{B} \subset \mathcal{L}(X)$ are γ -bounded, then $\mathcal{A}\mathcal{B} = \{AB : A \in \mathcal{A}, B \in \mathcal{B}\}$ is γ -bounded and*

$$\gamma(\mathcal{A}\mathcal{B}) \leq \gamma(\mathcal{A})\gamma(\mathcal{B}).$$

Proof. For finite choices A_j, B_j, x_j , apply the defining estimate first to A_j with input $B_j x_j$, and then to B_j with input x_j . $\square$

4 The full answer

Theorem 3. *Let $(T_t)_{t \geq 0}$ be a semigroup on an arbitrary Banach space X , and suppose*

$$M := \sup_{t \geq 0} \|T_t\| < \infty.$$

The following assertions are equivalent:

1. $\{T_s : 0 \leq s < 1\}$ is γ -bounded;
2. $\{e^{-\delta_0 t} T_t : t \geq 0\}$ is γ -bounded for some $\delta_0 > 0$;
3. $\{e^{-\delta t} T_t : t \geq 0\}$ is γ -bounded for every $\delta > 0$.

More quantitatively, if the family in (2) has γ -bound G_0 , then for every $\delta > 0$,

$$\gamma(\{e^{-\delta t} T_t : t \geq 0\}) \leq \frac{M e^{(\delta_0 - \delta)_+}}{1 - e^{-\delta}} G_0, \quad (r)_+ := \max\{r, 0\}. \quad (1)$$

Proof. The implication (3) $\Rightarrow$ (2) is immediate. If (2) holds, then on $0 \leq s < 1$,

$$T_s = e^{\delta_0 s} (e^{-\delta_0 s} T_s).$$

The Gaussian contraction principle therefore gives (1).

Assume (1), fix $\delta > 0$, and set

$$\mathcal{A}_\delta = \{e^{-\delta n}T_n : n = 0, 1, 2, \dots\}, \quad \mathcal{B}_\delta = \{e^{-\delta s}T_s : 0 \leq s < 1\}.$$

Lemma 1 and the boundedness of the semigroup give

$$\gamma(\mathcal{A}_\delta) \leq \sum_{n=0}^{\infty} e^{-\delta n} \|T_n\| \leq \frac{M}{1 - e^{-\delta}}.$$

The contraction principle shows that $\mathcal{B}_\delta$ is γ -bounded. Every $t \geq 0$ has a unique decomposition $t = n + s$ with $n \in \mathbb{N}_0$ and $0 \leq s < 1$, and the semigroup law yields

$$e^{-\delta t}T_t = (e^{-\delta n}T_n)(e^{-\delta s}T_s) \in \mathcal{A}_\delta \mathcal{B}_\delta.$$

Lemma 2 proves (3).

Finally, under (2),

$$e^{-\delta s}T_s = e^{(\delta_0 - \delta)s}(e^{-\delta_0 s}T_s), \quad 0 \leq s < 1.$$

Thus $\gamma(\mathcal{B}_\delta) \leq e^{(\delta_0 - \delta)G_0}$, which combined with the preceding discrete estimate proves (1). $\square$

Corollary 4 (answer to Question 5.8). *The K -convexity assumption in Corollary 5.5 of arXiv:1907.03113 can be dropped completely.*

Proof. The hypothesis of that corollary is assertion (2) of Theorem 3, and its conclusion is assertion (3). $\square$

5 Upgrade: equality of growth bounds

Suppose $-A$ generates a C_0 -semigroup (T_t) , and use the source's sign convention

$$\omega(A) = \sup\{b : \sup_{t \geq 0} e^{bt} \|T_t\| < \infty\}, \quad \omega^\gamma(A) = \sup\{a : \{e^{at}T_t : t \geq 0\} \text{ is } \gamma\text{-bounded}\}.$$

Corollary 5. *On every Banach space, if (T_t) admits at least one γ -type, then*

$$\omega^\gamma(A) = \omega(A).$$

Proof. Always $\omega^\gamma(A) \leq \omega(A)$. For the reverse inequality, fix $\beta < \omega(A)$. Choose b with $\beta < b < \omega(A)$. By definition, $S_t = e^{bt}T_t$ is a bounded semigroup. Let a_0 be any γ -type of (T_t) and choose $a < \min\{a_0, \beta\}$. The contraction principle shows that a is also a γ -type, so

$$\{e^{at}T_t : t \geq 0\} = \{e^{-(b-a)t}S_t : t \geq 0\}$$

is γ -bounded, with $b - a > 0$. Apply Theorem 3 to the bounded semigroup (S_t) at the new damping $b - \beta > 0$. It follows that

$$\{e^{-(b-\beta)t}S_t : t \geq 0\} = \{e^{\beta t}T_t : t \geq 0\}$$

is γ -bounded. Hence every $\beta < \omega(A)$ is a γ -type, which proves the claimed equality. $\square$

6 Scope and novelty

No continuity, K -convexity, finite cotype, or property (α) enters Theorem 3; apart from boundedness, only the semigroup identity is used. Corollary 5 removes K -convexity from the equality $\omega^\gamma(A) = \omega(A)$ when a γ -type exists. It does *not* identify the resolvent abscissa $s_0^\gamma(A)$ and hence does not answer the source's separate Question 5.9.

A bounded search of the run indexes, the exact question, the source title, and the displayed damped-orbit formula found no later explicit answer. The source proves only the uniformly continuous case in Remark 5.6. Because the argument is elementary, an unindexed folklore antecedent remains possible; the mathematical claim is therefore submitted as a candidate full solution with moderate novelty confidence.

7 Verification and review recommendation

The proof is analytic and uses no computational evidence. The source crop was rendered directly from the official arXiv PDF. The LaTeX packet was compiled with all references resolved, its log was checked for layout warnings, and all rendered pages were visually inspected. Expert review is recommended, focusing on the conditional-expectation step for repeated indices and on the sign choices in Corollary 5. Subject to that review, the packet should be treated as a full affirmative solution to Question 5.8; Question 5.9 remains outside its scope.

References

- [1] Loris Arnold, *γ -boundedness of C_0 -semigroups and their H^∞ -functional calculi*, arXiv:1907.03113v1, 2019.

of γ -type s for any $s < s_0^\gamma(A)$. Thus $\omega^\gamma(A) \geq s_0^\gamma(A)$. Combining with (24), we obtain the result. $\square$

EXAMPLE 5.4. Let (Ω, μ) be a measure space and let $1 < p < \infty$. According to [1, Theorem 5.3.6], if $-A$ is the generator of a *positive* C_0 -semigroup $(T_t)_{t \geq 0}$ on $L^p(\Omega)$, then $s_0(A) = \omega(A)$. If in addition $\omega^\gamma(A) > -\infty$ then the equalities $\omega^\gamma(A) = \omega(A) = s_0^\gamma(A) = s_0(A)$ hold by Corollary 5.3.

The equality $\omega^\gamma(A) = \omega(A)$ in Corollary 5.3 implies the following statement.

COROLLARY 5.5. *Let $(T_t)_{t \geq 0}$ is a bounded C_0 -semigroup on some K -convex Banach space. If there exists $\delta > 0$ such that $\{e^{-\delta t}T_t : t \geq 0\}$ is γ -bounded, then $\{e^{-\delta t}T_t : t \geq 0\}$ is γ -bounded for any $\delta > 0$.*

REMARK 5.6. Let $(T_t)_{t \geq 0}$ is a bounded C_0 -semigroup on X , with generator $-A$. If A is bounded (equivalently, if $(T_t)_{t \geq 0}$ is uniformly continuous), then the property considered in the above statement is true, that is, $\{e^{-\delta t}T_t, t \geq 0\}$ is γ -bounded for any $\delta > 0$.

Indeed, consider $\delta > 0$. Since A is bounded and $\sigma(A) \subset \overline{R_0}$, there exists an open disk D such that $\sigma(A + \delta) \subset D \subset R_0$. Let ∂D be the boundary of D oriented counterclockwise. Then by the Dunford-Riesz calculus, we have

$$e^{-\delta t}T_t = \frac{1}{2\pi i} \int_{\partial D} e^{-t\lambda} R(\lambda, A + \delta) d\lambda$$

for any $t \geq 0$. Then a straightforward application of Lemma 2.12 shows that $\{e^{-\delta t}T_t : t \geq 0\}$ is γ -bounded.

We conclude this section with an observation and two questions. First we state a result that we recently obtained (with C. Le Merdy).

THEOREM 5.7. ([2, Corollary 0.5]) *Let X be isomorphic to a separable Banach lattice with finite cotype such that X is not isomorphic to a Hilbert space. Then there exists $A \in \mathcal{L}(X)$ such that $\{e^{-tA} : t \geq 0\}$ is bounded but not γ -bounded.*

Combining this theorem with Remark 5.6, we obtain that Corollary 5.5 is sharp in the class of uniformly continuous semigroups. Namely on any K -convex separable Banach lattice not isomorphic to a Hilbert space (on L^p for $1 < p \neq 2 < \infty$, say) we obtain a uniformly continuous semigroup $(T_t)_{t \geq 0}$ such that $\{e^{-\delta t}T_t : t \geq 0\}$ is γ -bounded for any $\delta > 0$, $\{T_t : t \geq 0\}$ is bounded but $\{T_t : t \geq 0\}$ is not γ -bounded.

The assumption that X is K -convex space in Corollary 5.5 is quite surprising. This leads to the following question:

QUESTION 5.8. Does the assumption X is K -convex space in Corollary 5.5 can be dropped?

We recall (24) and the existence of $-A$ generating a C_0 -semigroup $(T_t)_{t \geq 0}$ such that $\omega(A) < s_0(A)$. So we ask

QUESTION 5.9. Does there exist an operator A such that $-A$ which generates a C_0 -semigroup $(T_t)_{t \geq 0}$, satisfying $\omega(A) < s_0^\gamma(A)$?

Figure 1: Corollary 5.5, the uniformly continuous special case, and Question 5.8 on page 20 of arXiv:1907.03113v1.